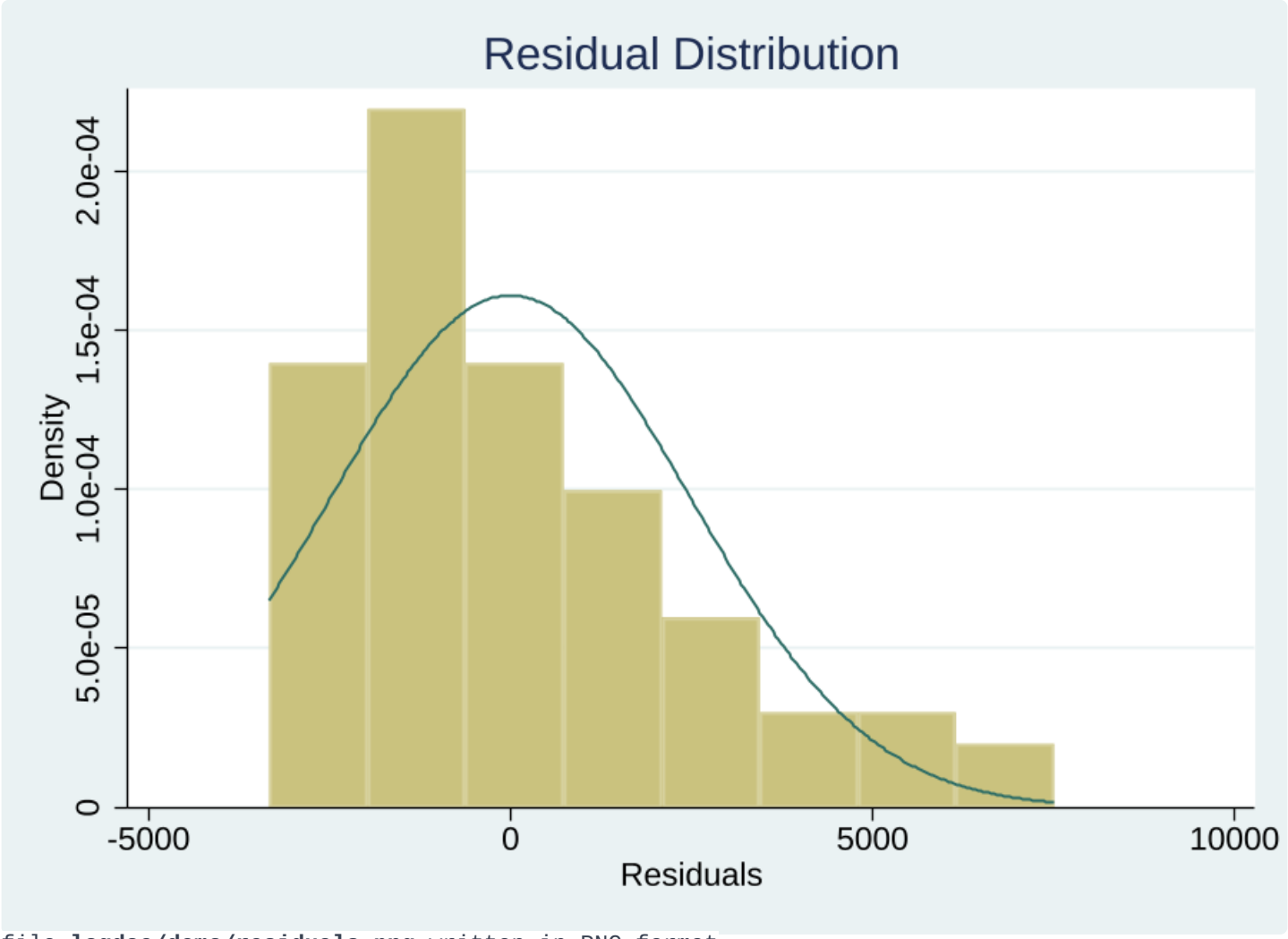
PDF Dependency Check

```
. * # Data Overview
. summarize price mpg weight length, separator(0)
   Variable |        Obs        Mean    Std. dev.       Min        Max
------------+---------------------------------------------------------
      price |         74    6165.257    2949.496       3291      15906
        mpg |         74     21.2973    5.785503         12         41
     weight |         74    3019.459    777.1936       1760       4840
     length |         74    187.9324    22.26634        142        233

. tabstat price mpg weight, by(foreign) statistics(n mean sd median min max) columns(statistics)
Summary for variables: price mpg weight
Group variable: foreign (Car origin)

foreign |         N      Mean        SD       p50       Min       Max
---------+--------------------------------------------------------------
Domestic |        52  6072.423  3097.104    4782.5      3291     15906
        |        52  19.82692  4.743297        19        12        34
        |        52  3317.115  695.3637      3360      1800      4840
---------+--------------------------------------------------------------
Foreign |        22  6384.682  2621.915      5759      3748     12990
        |        22  24.77273  6.611187      24.5        14        41
        |        22  2315.909  433.0035      2180      1760      3420
---------+--------------------------------------------------------------
  Total |        74  6165.257  2949.496    5006.5      3291     15906
        |        74   21.2973  5.785503        20        12        41
        |        74  3019.459  777.1936      3190      1760      4840
---------+--------------------------------------------------------------

. * # Regression Analysis
. regress price mpg weight length i.foreign
     Source |       SS           df       MS      Number of obs   =        74
-------------+----------------------------------   F(4, 69)        =     21.01
      Model |   348708940         4    87177235   Prob > F        =    0.0000
   Residual |   286356456        69  4150093.57   R-squared       =    0.5491
-------------+----------------------------------   Adj R-squared   =    0.5230
      Total |   635065396        73  8699525.97   Root MSE        =    2037.2

-------------+------------------------------------------------------------------
      price | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+------------------------------------------------------------------
        mpg |  -13.40719   72.10761    -0.19   0.853    -157.2579    130.4436
     weight |   5.716181   1.016095     5.63   0.000     3.689127    7.743235
     length |  -92.48018    33.5912    -2.75   0.008    -159.4928   -25.46758
            |
    foreign |
   Foreign  |   3550.194   655.4564     5.42   0.000     2242.594    4857.793
      _cons |    5515.58   5241.941     1.05   0.296    -4941.807    15972.97
-------------+------------------------------------------------------------------

. estimates store base
. * ## Margins
. margins foreign, atmeans

Adjusted predictions                                  Number of obs = 74
```

```
Model VCE: OLS

Expression: Linear prediction, predict()
At: mpg       =  21.2973 (mean)
   weight    = 3019.459 (mean)
   length    = 187.9324 (mean)
   0.foreign = .7027027 (mean)
   1.foreign = .2972973 (mean)

-------------+------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+------------------------------------------------------------
    foreign |
  Domestic  |   5109.794   306.6837    16.66   0.000     4497.977    5721.611
   Foreign  |   8659.987   517.9058    16.72   0.000     7626.794    9693.181
-------------+------------------------------------------------------------

. * # Tabulations
. tabulate foreign rep78, row
+----------------+
| Key            |
+----------------+
|   frequency    |
| row percentage |
+----------------+

           |                  Repair record 1978
Car origin |         1          2          3          4          5 |     Total
-----------+-------------------------------------------------------+----------
 Domestic |         2          8         27          9          2 |        48
          |      4.17      16.67      56.25      18.75       4.17 |    100.00
-----------+-------------------------------------------------------+----------
  Foreign |         0          0          3          9          9 |        21
          |      0.00       0.00      14.29      42.86      42.86 |    100.00
-----------+-------------------------------------------------------+----------
    Total |         2          8         30         18         11 |        69
          |      2.90      11.59      43.48      26.09      15.94 |    100.00

. * # Diagnostics
. quietly regress price mpg weight
. predict double resid, residuals
. histogram resid, normal ///
>     title("Residual Distribution") ///
>     xtitle("Residuals") ytitle("Density")
(bin=8, start=-3332.4617, width=1354.9264)

. graph export "logdoc/demo/residuals.png", replace width(900)
```




```
file logdoc/demo/residuals.png written in PNG format

. capture graph close _all
. drop resid
```